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SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

Tutorial MS Azure

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Table Of Content

Project Overview.....	3
Description on the solution.....	3
Technology Use.....	3
Data Pipeline Architecture.....	3
MS Power BI dashboard.....	3
Reflection.....	4

Project Overview

This project addresses a critical gap in understanding customer demographics, specifically how gender distribution influences product purchases. Currently, the company maintains its customer records in a local SQL database, which makes it difficult for stakeholders to get a clear, real-time view of these trends. To solve this, the company has requested a comprehensive KPI dashboard that tracks total products sold, total sales revenue, and the gender split across different categories.

Our solution involves building an automatic system that moves all that data into Microsoft Azure. We extract the information from the local database and organize it using three simple steps: raw, cleaned, and ready-to-use. This makes sure the data is accurate and easy to query. The best part is that the system updates itself every day. This means the team always sees the latest numbers without having to do any manual work.

We also made sure the project is safe and can grow with the company. We use special Azure tools to hide passwords and control who can see the data to keep everything secure. The system is also built to handle a huge amount of information, so it will still work perfectly even if the business gets much bigger in the future.

The final result is an easy Power BI dashboard that shows sales and customer trends through simple charts. It includes a user-friendly interface for date-based queries and allows stakeholders to filter by category or gender. Basically, this project takes data that was hard to get to and turns it into a helpful tool. It helps the company understand its market and make better plans for the future.

Description on the solution

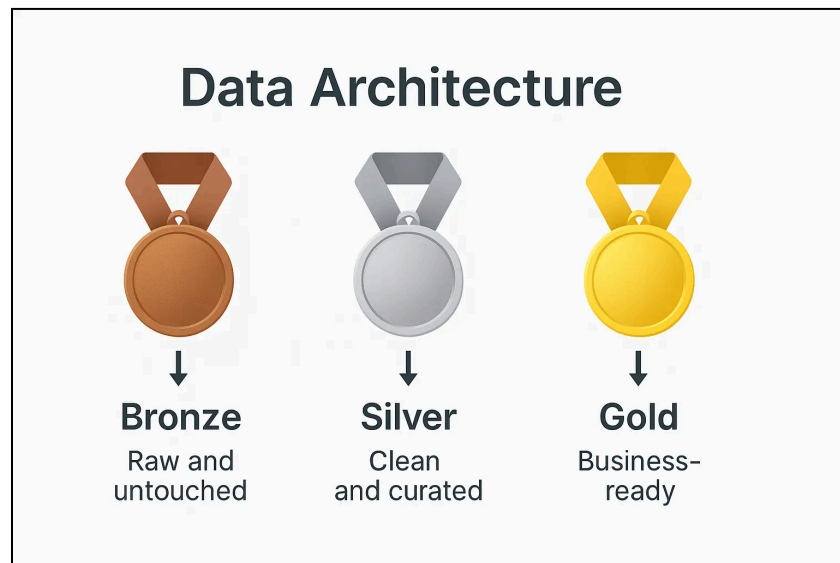


Figure 1: The Medallion Architecture

The solution is built around a "Medallion Architecture" as in *Figure 1* which is a smart way to manage data quality as it flows through the pipeline. Instead of just dumping everything into one place, this structure organizes the data into three distinct layers to make sure it is perfectly refined and ready for the business to analyze.

The process starts at the Bronze Layer, which acts as the first landing zone for all the information coming in from the local SQL database. At this stage, we keep the data exactly as it is in its original, raw format without any changes. Having this "source of truth" is really important because it gives the team a safety net if something goes wrong later on, we can always go back to the original files and start again.

Once the raw data is safe, it moves into the Silver Layer for some much-needed cleaning. This is where we use Azure Databricks to tidy things up and make the data more consistent. For instance, we take messy or mismatched date formats and turn them into one clean, standard style. We also make sure all the columns are clear and easy to read, which makes the data a lot more reliable for the final steps.

Finally, the data reaches the Gold Layer, where it is fully polished and ready for reporting. Here, we focus on making the information as "query-friendly" as possible, such as using snake_case for all the column names. These refined tables are specifically prepared so that Power BI can easily pick them up and turn them into the helpful charts and business insights the team needs to make better decisions.

Technology Use

Introduction Microsoft Azure represents one of the most widely deployed cloud computing platforms, providing users with several cloud services focused on big data and contemporary data engineering. At the current Tutorial level, Azure provides solutions that are extremely scalable, elastic, and safe. The integration of its ecosystem allows moving from fragmented monitoring approaches to comprehensive automation and pipeline creation. To implement the present data engineering task, four Azure services are utilized: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, and Azure Data Lake Storage Gen2 (ADLS Gen2).

SQL Server Management Studio (SSMS)

The role of the SQL Server Management Studio is limited to management of the source environment. In the current architecture, it is applied to control operations with on-premises and virtualized SQL databases where the source datasets reside. As a result, before any cloud migration efforts take place, it is used to evaluate the quality of data and prepare queries needed to understand which exact datasets should be ingested into the Azure cloud environment.

Azure Data Factory (ADF)

This service represents the orchestration engine of the whole pipeline. It does not do much processing on its own but rather coordinates all other processes, including migration and transfer of the data from the source to the cloud. Using the Integration Run Times as connectors, ADF triggers the process of moving data (Copy Activity) from the source server(s) (which are found via SSMS) to the storage layer.

Azure Data Lake Storage Gen2 (ADLS Gen2)

The Azure Data Lake Storage Gen2 represents the foundational storage of our system that utilizes Medallion architecture. The structure of ADLS consists of hierarchical namespaces used to represent different zones of data (Bronze, Silver, Gold). Thus, ADLS provides an efficient way of storing massive amounts of semi-structured data as the main storage of the organization with extreme reliability and safety. In other words, it is the "single source of truth" for our enterprise.

Azure Databricks

In order to process huge amounts of structured and semi-structured data in the Azure, we require Azure Databricks (the platform built on top of the Apache Spark framework). It represents the processing part of our pipeline. By applying complex business rules, it filters out duplicates, handles null values, etc. With the help of Databricks, the source raw data collected by ADF is transformed into the curated form required for the next step.

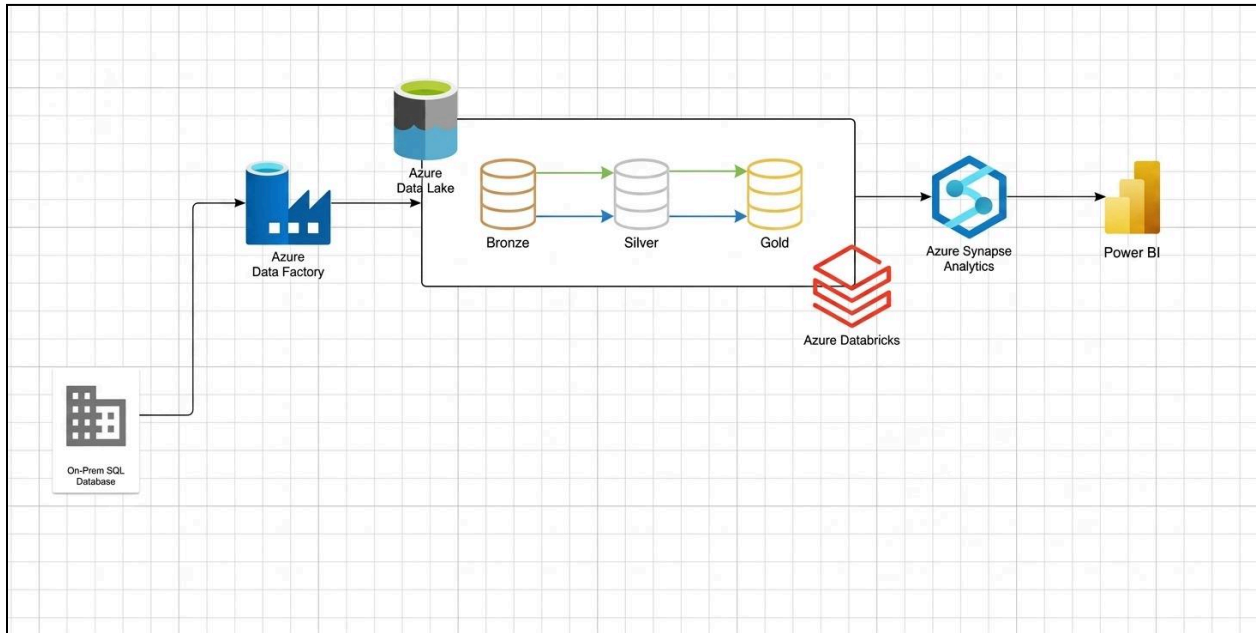
Azure Synapse Analytics

Azure Synapse Analytics represents the enterprise data warehouse used in our architecture. It receives data after the transformations applied in the previous step and loads it into Dedicated SQL pools. Synapse is optimized to handle complex OLAP queries against massive relational datasets. In other words, it bridges the gap between the flexibility of a data lake and the need for a structured database.

Power BI

Power BI represents the visualization part of the whole architecture used to create dashboards and reports. Power BI directly connects to Azure Synapse Analytics to get data. Since we already did all the data cleansing at the previous steps, this service can immediately display all required information with low latency.

Data Pipeline Architecture



Based on the figure above, the data pipeline used for this tutorial follows a Medallion Architecture to move the data from on-premises SQL database to an interactive dashboard using Microsoft Azure services. The process begins with Azure Data Factory (ADF) pulling data from SQL databases through connection to ingest the raw data from AdventureworksLT2022. This raw data is first transported into the Bronze layer of an Azure Data Lake Storage Gen2 (ADLS Gen2) file containers. From there, Azure Databricks performs transformations using Apache Spark and its own compute cluster to clean the data and moves it into the Silver layer. Databricks then applies further business logic and renames columns to prepare the data for the Gold layer, where it exists as business-ready aggregates. Once refined, the data is loaded into Azure Synapse Analytics, which serves as the enterprise data warehouse optimized for complex analytical queries. Finally, Power BI connects directly to Synapse to pull this structured data, translating it into interactive dashboards for real-time stakeholder insights. The entire pipeline is secured using Azure Key Vault to manage credentials and Azure Active Directory for role-based access control.

MS Power BI dashboard

To transform raw data into clear and actionable stories, three specialized Power BI dashboards have been developed. As shown in Figure 1.1, the first dashboard analyzes customer purchasing behavior, specifically identifying patterns in product selection and spending habits based on gender. This level of detail is critical for marketing strategies.

The second dashboard, shown in Figure 1.2, evaluates sales team performance by providing a comprehensive view of individual achievements and overall team efficiency.

The final dashboard in Figure 1.3 illustrates the company’s market footprint, utilizing city and regional data to pinpoint high-growth areas and geographic sales density.

Overall, these analytical tools provide the deep insights necessary to drive informed, data-driven decision-making and support the company’s strategic growth.

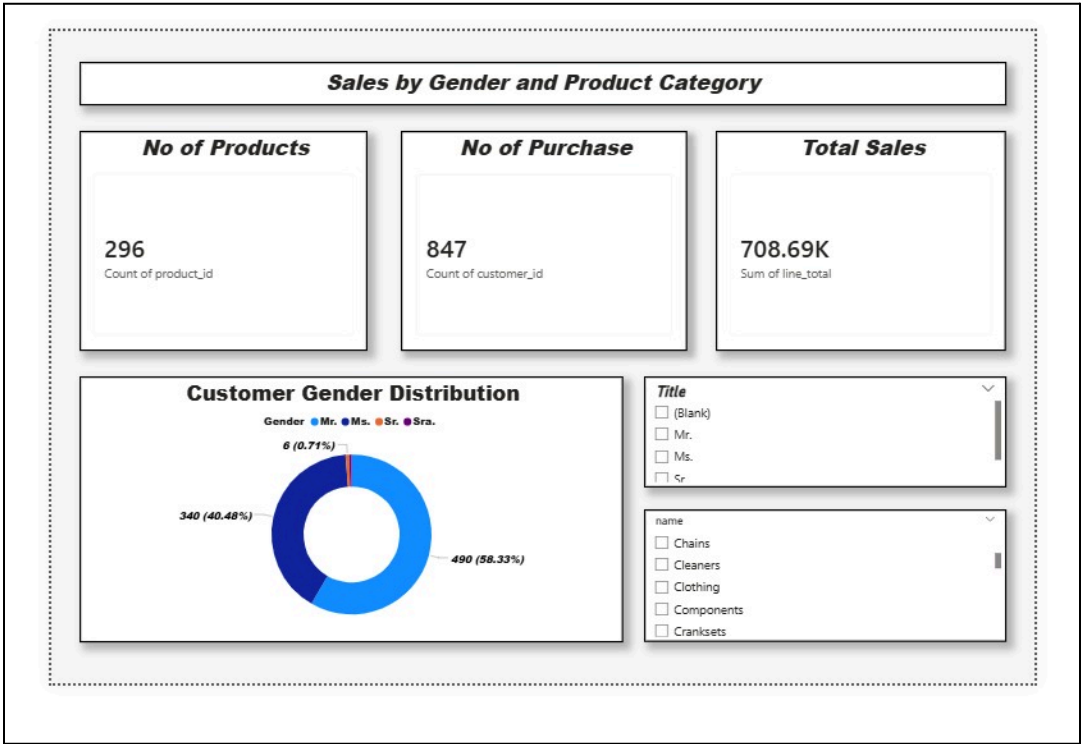


Figure 1.1 Sales By Gender and Product Category



Figure 1.2 Sales Person Performance

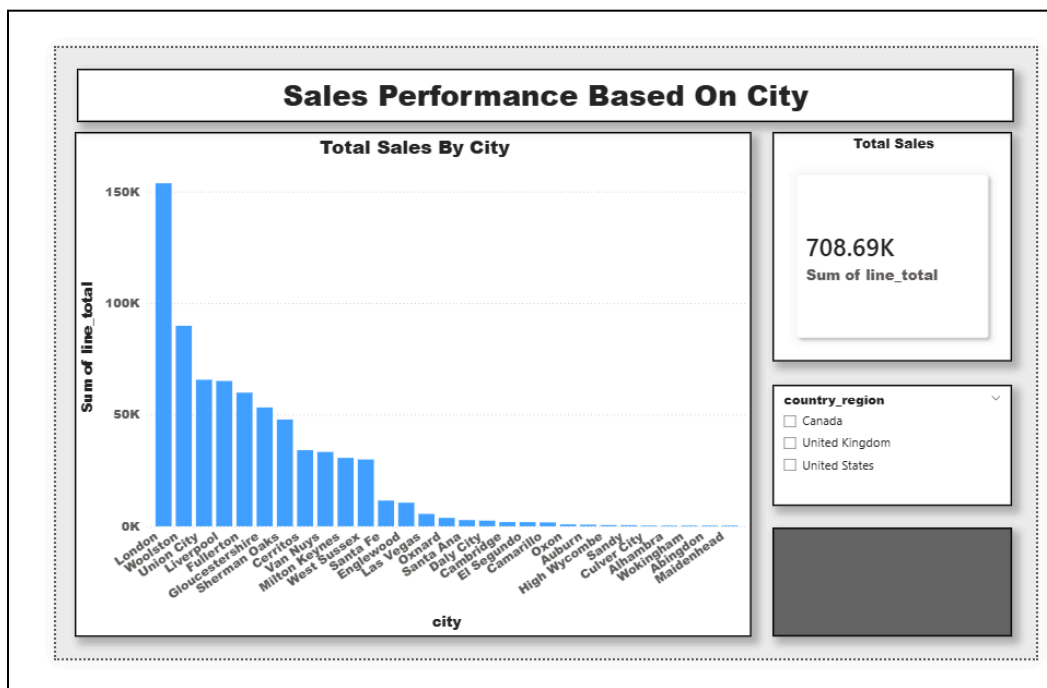


Figure 1.3 Sales Performance Based On City

Reflection

Muhammad Naim Bin Abdullah

Throughout this project, I learn so many new things that I am not familiar with or maybe forgot about. The most impactful thing that I learned is how to create a data pipeline by using azure. This shows me how it works and also lets me get some kind of idea on how data engineering does. Even though there are a few challenges that I face in this tutorial, I still keep on going until the end because I found it really interesting and exciting. Overall, this tutorial has made me more confident to make a data pipeline by myself.

Muhammad Adam Bin Razali

Before taking this tutorial, I do not really understand the real application for a data pipeline for oltp and olap data. It really makes it clear what tools from cloud services are useful in developing a cloud-based data pipeline. Although I encountered lots of problems and mistakes during the setup of the pipeline, I always find a way to make it work and eventually learning the concept better from all those mistakes. Furthermore, I also learn how to express the data into an actual business insights through real-time analytic dashboard by turning the raw data into a meaningful data for data-driven business decision.

Afif Shaqir Irfan Bin Arqam

Before this project, terms like "data pipelines" were just words for me. Actually building one made it much easier for me to understand what it really means. The project had some difficult moments. The setup was one of the hardest parts, as connecting a local database to the cloud was not easy. But solving these problems ended up being the most useful part of the learning process. One of the best parts was seeing messy, raw data turn into a clean and simple dashboard in Power BI. It showed how important the background work is in helping people make good decisions. Most of all, this project helped build confidence in managing a data project from start to finish. It showed that even hard problems can be solved with patience and a clear approach.